1. Team Name

SYNDICATE-32

2. Project Name

Grid-Connected DC Microgrid with Net Metering.

3. Introduction

A DC microgrid is a compact power distribution system that operates on direct current and integrates renewable sources such as solar and wind. Our project demonstrates a grid-connected DC microgrid with intelligent control, energy storage, and load prioritization. The system automatically switches to battery backup during grid failure, ensuring that essential loads continue operating while non-essential loads are shed. Additionally, the microgrid features a net-metering mechanism, allowing excess renewable energy to be fed back to the grid once the battery energy storage system is fully charged. This prototype highlights how smart microgrids improve reliability, resilience, energy efficiency, and renewable integration.

4. Motivation

Bangladesh experiences fluctuations in power availability, especially in coastal, island and rural regions. Critical infrastructures such as hospitals, communication networks and traffic systems require stable and uninterrupted electricity. A grid-connected DC microgrid provides a realistic solution by combining renewable generation, energy storage, and intelligent load management. With net metering, excess energy can be fed back to the grid, reducing dependence on national supply and lowering costs. This project builds an educational, scalable demonstration of such a system.

5. Objective

- To design and implement a grid-connected DC microgrid with net metering capability.
- To integrate solar and wind as renewable inputs.
- To enable automatic switching between grid power and battery storage.
- To emulate real-world essential and non-essential load profiles for demonstration.
- To display battery level, operating mode, and backup time via IoT.

6. Components

<u>Sensors & Monitoring:</u>

- ACS712 current sensor
- Voltage sensor
- 16×2 LCD displays
- LDR Sensor

Power & Control:

- Solar panel (90×90 mm, 5V, 200 mA)
- 5V DC Motor
- Buck/boost converters
- 3S battery pack
- Relay module (active low, 4-channel)
- NodeMCU ESP8266
- Arduino Uno
- Arduino Nano
- Logic level shifter

Load Emulation:

- LEDs
- COB LEDs
- Buzzer

Miscellaneous:

- Wires, PVC board, mounting accessories

7. Social Impacts

- Supports critical services during outages by prioritizing essential loads.
- Promotes renewable energy, reducing dependence on fossil fuels.
- Provides educational value, helping students and communities understand modern energy systems, IoT monitoring, net metering, and smart microgrid control.
- Scalable concept, applicable for micro-industries, smart homes and remote electrification like coastal areas or islands.

8. Cost Analysis

Item Category	Estimated Cost (BDT)
Sensors & Indicators	1290
Power & Control	3440
Load Emulation	145
Miscellaneous	410
Total Cost	5285

9. Output

- The microgrid successfully switches between grid mode and battery backup mode.
- During grid failure, only essential loads remain active.
- Renewable inputs (solar/wind) charge the battery when the grid is present.
- When the battery becomes fully charged, excess renewable energy is routed back to the grid through controlled relay switching.
- Load profiles (24/7, day-only, night-only, and non-essential) are demonstrated in real time.
- Display-1 shows Grid Status (Online / Offline), battery storage system percentage and voltage.
- One of the ACS712 sensor provides real-time power consumption, current and mode profiles on Display-2.
- Another ACS712 sensor provides real-time power consumption for net metering on Display-3.

10. Summary

This project presents a fully functional prototype of a grid-connected DC microgrid featuring renewable integration, load prioritization, battery storage, IoT monitoring, and net metering. The system ensures continuity of essential loads during grid failure while intelligently disconnecting non-essential loads to conserve battery power.

Real-time IoT monitoring through the NodeMCU web server provides battery status, operating mode, and backup estimates, making the system transparent and user-friendly. By incorporating net metering, the model showcases how renewable energy can be efficiently utilized and fed back into the grid, reducing energy costs and supporting sustainable development.

This scalable microgrid concept is suitable for island electrification, emergency backup systems, smart homes, and educational demonstrations.